

CS 5330 Pattern Recognition and Computer Vision

PROJECT PROPOSAL

GENERAL PROJECT INFORMATION

Project Name	Course Professor
Reproducing VoxelMorph: Unsupervised Learning for Multimodal Medical Image Registration.	Dr. Akram Bayat

Student Name	E-mail	Role
Mingzhe Ou	ou.mi@northeastern.edu	Algorithmic Lead
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1. PROJECT OVERVIEW

In this section, define the problem you aim to solve using computer vision techniques. Clearly explain why it is an important problem and what the real-world application of the solution will be.

Introduction	-Summary: This project reproduces VoxelMorph for unsupervised T1-to-T2 brain MRI registration and compares its performance against a classical mutual-information baseline. -Motivations: To overcome the high computational cost and intensity cross-modal challenges of traditional registration by using a fast, learning-based alignment. Project Type: Reproducing Project (<i>VoxelMorph</i>)
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Problem Statements	-Problem Description: Aligning multi-sequence (T1/T2) brain MRI scans to a unified coordinate system for anatomical comparison. -Context/Domain: Healthcare / Medical Imaging. Using the OASIS-1 dataset for neuroimaging analysis. -Challenges & Gaps: Traditional methods are computationally slow and struggle with cross-modal intensity differences. This project aims for rapid, unsupervised alignment without ground-truth labels.
Objective (Goals)	-Project Goals To reproduce VoxelMorph for fast, unsupervised T1-to-T2 brain MRI registration and validate its accuracy against classical methods. -Planned Outputs -Trained Model: An unsupervised network for cross-modal MRI alignment. -Classical Baseline: A SimpleITK script using Mutual Information for benchmarking. -Evaluation: A comparative report including Dice scores and 3D Slicer visualizations.
Deliverables	-GitHub Repository: Source code for VoxelMorph implementation, preprocessing scripts, and the SimpleITK baseline. -Trained Model: Optimized weights for unsupervised T1-to-T2 MRI registration. -Final Report: Comparative analysis featuring Dice scores, Jacobian maps, and 3D Slicer visualizations. -Data Pipeline: A standardized, preprocessed version of the OASIS-1 dataset.

2. DATA COLLECTION PLAN

Describe how you plan to collect the data or where you will source it from. Outline any preprocessing that will be necessary to make the data usable.

Data Source	The project utilizes the existing public OASIS-1 dataset, which contains paired T1 and T2 brain MRI scans. The data is accessed via official academic repositories or Kaggle.
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Dataset Size & Characteristics	-Size: A curated subset of 50 paired T1/T2 MRI scans from OASIS-1. -Resolution: Downsampled/cropped to $128 \times 128 \times 128$ for computational efficiency. -Preprocessing: Includes Min-Max normalization, N4 bias field correction, and Affine pre-alignment to standardize inputs.
DataCollection(If applicable) / Annotation	-Logistics: We use the existing OASIS-1 MRI volumes data so no custom data collection is required. -Annotation: No manual labeling like LabelMe or CVAT is needed since VoxelMorph is unsupervised learning framework. -Evaluation Ground Truth: For performance validation, the project will use the pre-segmented anatomical labels that are already built in OASIS-1 dataset to calculate Dice scores.

3. METHODS

Outline the specific computer vision techniques and algorithms you plan to use.

Proposed Approach	-Unsupervised Registration: Predicting deformation fields via VoxelMorph without labels. -Spatial Transformers (STN): Using differentiable warping for end-to-end learning. -Multi-Modal Loss: Optimizing via Mutual Information (MI) for T1/T2 alignment. -Classical Baseline: Benchmarking against SimpleITK iterative methods.
Algorithms/ Models	-Algorithms & Models Primary Model: VoxelMorph (U-Net based CNN). Core Layer: Spatial Transformer Network (STN) for warping. Classical Baseline: SimpleITK (Mattes Mutual Information). -Processing Steps 1. Preprocessing: Affine alignment and normalization. 2. Training: End-to-end optimization of Similarity and Smoothness loss. 3. Inference: Fast, single-pass generation of deformation fields.

	-Optimization Hardware: GPU Acceleration (PyTorch/CUDA) for 3D volumes. Tuning: Optimizing the regularization weight (λ) for field smoothness.
Software/Tools:	-Libraries: PyTorch (Deep Learning), MONAI (Medical Imaging), SimpleITK (Baseline). -Environment: Google Colab / Local GPU (NVIDIA CUDA).

4. EVALUATION PLAN

Explain how you plan to evaluate the performance of your application or model, and compare it with existing solutions. Define metrics and benchmarks.

Evaluation Metrics	-Dice Score: this assesses the overlap accuracy of the labels. -Jacobian Determinant: Check for smooth and how realistic the deformations are. -Baseline: Performance vs. SimpleITK (Classical Mutual Information).
Testing Plan (if applicable)	-Quantitative Test: checking dice scores on a test set. -Qualitative Test: Visualizing alignment through Checkerboard and Overlay plots.

5. PRESENTATION PLAN

Explain how your group will present your results, both visually and in writing. Include plans for a poster, a written report, and a short video.

Poster	-Framework Architecture: A diagram of the VoxelMorph (U-Net + STN) pipeline. -Methodology: Concise summary of Unsupervised Learning and Loss Functions. -Visual Results: Checkerboard and Overlay plots comparing T1/T2 alignment. -Quantitative Analysis: Graphs of Dice Scores and Jacobian Determinants vs. Baseline.
GitHub Repo	- https://github.com/its-mz/CS5330_SP26_Group6.git

Write Up	-Format: Adhering strictly to IEEE 2-column conference template. -Sections: Standard structure including Intro, Methods, Experiments, and Conclusion. -Team Work: Each member will draft their respective technical modules and perform mutual peer review.
Video Clip	- N/A for this phase

6. TENTATIVE SCHEDULE

Provide a brief timeline, allocating approximate dates for each stage of the project.

KEY MILESTONE	START	FINISH
Form Project Team	01/26/2026	02/13/2026
Finalize Project Plan / Kick Off	02/14/2026	02/22/2026
Define Phase (e.g., Problem Definition, Dataset, Pre-processing, initial setting)	02/23/2026	03/09/2026
Development Phase (develop and test initial model)	03/10/2026	03/30/2026
Analysis Phase (model tuning, optimization, and evaluation)	03/31/2026	04/05/2026
Control Phase (finalize results)	04/04/2026	04/13/2026
Presentation Phase	04/04/2026	04/23/2026

REFERENCES

1. G. Balakrishnan, A. Zhao, M. R. Sabuncu, J. Guttag, and A. V. Dalca, "voxelmorph," GitHub. [Online]. Available: <https://github.com/voxelmorph/voxelmorph>. [Accessed: Mar. 09, 2026].